AI-BASED SMART SURVEILLANCE SYSTEM

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Summary: (Min 50 - Max 100 words)	This project presents an AI-Based Smart Surveillance System that combines real-time face recognition, weapon detection, automated access control, and cloud monitoring. The system is implemented on a Raspberry Pi 4 using Dlib CNN-based face recognition and YOLOv8 object detection. Access is granted only when an authorized face is recognized and no weapon is detected. Firebase integration enables real-time notifications, event logging, and remote monitoring through a Flutter mobile application. The system achieved 97% face recognition accuracy and 93% weapon detection accuracy while providing a low-cost and scalable smart security solution.				
Introduction (One line)	An AI-powered surveillance and access control system that enhances security through real-time face recognition, weapon detection, and automated door control.				
Problem Statement (Max 2 lines)	Traditional CCTV and access control systems rely heavily on human monitoring and are vulnerable to unauthorized access. An intelligent, automated, and low-cost security solution capable of real-time threat detection and access control is required.				
Objective	To design, develop, and validate an AI-powered surveillance system capable of real-time face recognition, weapon detection, automated door access control, and cloud-based remote monitoring.				
Result(s) & Conclusions	The system achieved **97% face recognition accuracy** and **93% weapon detection accuracy** with reliable real-time operation on Raspberry Pi 4. Integrated testing confirmed successful automated access control, Firebase synchronization, mobile notifications, and stable hardware performance, demonstrating the feasibility of an intelligent low-cost surveillance solution.				
Future Directions	➢ ➤ Facial anti-spoofing for enhanced security ➤ Multi-camera surveillance support ➤ Advanced behavior and threat analysis ➤ Solar-powered operation for energy efficiency ➤ Cloud-based analytics and reporting dashboard ➤ Large-scale deployment with centralized management				



